

minilab3/minilab3-fixed.c

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <pthread.h>
4
5  int sharedCounter = 0;
6  int num_iterations;
7  pthread_mutex_t mutex;
8
9  void *thread(void *vargp)
10 {
11     for (int i = 0; i < num_iterations; i++) {
12         pthread_mutex_lock(&mutex);
13         sharedCounter++;
14         pthread_mutex_unlock(&mutex);
15     }
16 }
17
18 int main(int argc, char *argv[]) {
19     if (argc != 3) {
20         printf("Usage: %s <#_iteration> <#_thread>\n", argv[0]);
21         exit(1);
22     }
23
24     num_iterations = atoi(argv[1]);
25     int num_threads = atoi(argv[2]);
26
27     if (num_threads > 10) {
28         printf("Maximal number of threads is 10\n");
29         exit(1);
30     }
31
32     pthread_mutex_init(&mutex, NULL);
33     pthread_t tids[num_threads];
34
35     for (int i = 0; i < num_threads; i++) {
36         pthread_create(&tids[i], NULL, thread, NULL);
37     }
38     for (int i = 0; i < num_threads; i++) {
39         pthread_join(tids[i], NULL);
```

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40     }
41
42     printf("Threads: %d, Expected Iterations: %d, Actual Iterations: %d\n",
43           num_threads, num_iterations * num_threads, sharedCounter);
44     pthread_mutex_destroy(&mutex);
45     exit(0);
46 }
47
```